

Inference and Observational Learning under Common Information: Relative Confidence and Its Calibration*

Andrea Amelio[†]

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Abstract

People who observe the same statistical evidence can nevertheless reach different conclusions because they differ in inference quality. This creates scope for beneficial social learning even when another person possesses no additional raw information. We study this possibility in a balls-and-urns experiment in which a participant and a peer face the same prior, signal structure, and realized signal. After reporting an initial posterior, participants observe the peer’s posterior and may revise. Final beliefs are more accurate on average after peer observation. Relative cognitive uncertainty—the participant’s uncertainty about their own reasoning relative to the uncertainty attributed to the peer—is aligned with objective relative accuracy. Participants also place more weight on the peer when relative confidence favors the peer, and objectively better peer posteriors receive more than twice as much weight as worse ones. A compound-inference block provides further evidence for this account: when participants become more uncertain relative to a peer solving the corresponding reduced problem, they rely substantially more on the peer, learning gains increase, and peer quality becomes correspondingly more important for final accuracy. The results show that common information does not eliminate the value of social learning: heterogeneous inference creates gains from observing others, and the calibration of relative confidence determines whether those gains are realized.

Keywords: beliefs, observational learning, inference, relative confidence, cognitive uncertainty

JEL: C91, D83, D91

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[†]Bocconi University and IGIER: andrea.amelio@unibocconi.it

1 Introduction

Economic decisions are often made from the same statistical information. Financial analysts read the same earnings releases, policymakers inspect the same economic statistics, and experts evaluate the same evidence. Yet common information does not imply common conclusions. Even when an inference problem has a unique normative solution, people differ in how accurately they process and combine the information they observe.

This heterogeneity creates a simple but important possibility: people may learn from one another even when nobody has additional raw information. If two perfect Bayesian agents share the same prior, model, and signals, one agent’s posterior is redundant for the other. With imperfect inference, however, two people can produce estimates of different quality from the same evidence. Another person’s posterior is then a second computation of the same statistical object, and observing it can improve decisions.

Recent work establishes this broad point. Agranov et al. (2026) show that observing others’ actions can improve choices even when the underlying information is public, while Offerman and Sonnemans (1998) show that imperfect Bayesian agents can benefit from observing actions based on the same observed signals. We build on this idea in a canonical balls-and-urns experiment. Rather than observing a binary action, participants observe another person’s posterior from exactly the same balls-and-urns problem. Using the Bayesian posterior as a benchmark, we can measure the accuracy of both inferences and ask whether social observation helps, and how this connects to our mechanism.

Individuals recognize that both their own inference and that of others may be imperfect. They form subjective assessments of the reliability of their own reasoning and, after observing a peer, of the reliability of the peer’s reasoning. Relative confidence summarizes these two assessments and regulates the extent to which the peer’s inference affects the individual’s posterior. Whether this process improves accuracy depends on calibration: when perceived relative reliability is aligned with relative performance, social learning is beneficial; when it is not, revision can reduce accuracy. This idea connects to a broader principle developed and presented by (Enke et al., 2023): institutions that aggregate individual decisions, such as markets, perform better when individuals are well-calibrated about their own abilities. The same logic applies to social learning: whether observing others improves or worsens decisions depends on whether people’s relative confidence is aligned with their relative performance (Amelio, 2023).

We examine these ideas in an online experiment with two phases. Phase 1 generates pools of posterior estimates for a set of Bayesian balls-and-urns problems. In Phase 2, participants solve the same problems, report an initial posterior and cognitive uncertainty about their reasoning, observe a randomly selected peer posterior from the matching Phase-1 problem, evaluate the peer, and may revise. In the baseline block, participant and peer face exactly the same prior, signal-generating process, and realized signal. The peer therefore contributes no private signal. The

experiment gives us an objective Bayesian benchmark, a measure of relative confidence, a measure of how much weight the peer receives, and a realized change in accuracy after social observation.

The results form a simple sequence. First, observational learning is beneficial on average. Across baseline trials, final beliefs are 0.415 percentage points closer to the Bayesian benchmark than initial beliefs; among cases in which participant and peer initially disagree, the gain is 0.500 points. The gain is strongly related to peer quality: participants gain 1.41 points when the peer is more accurate and lose 0.47 points when the peer is less accurate. Thus, even with identical information, another person’s inference can improve accuracy because inference quality differs across people.

Second, relative confidence helps explain why these gains are positive. We define relative cognitive uncertainty (CU henceforth) as the participant’s uncertainty about their own reasoning minus the uncertainty attributed to the peer. Relative CU predicts objective relative accuracy: its sign identifies the more accurate posterior in 59.6% of non-tied baseline cases, significantly above chance. It also predicts reliance on the peer. A ten-point increase in relative CU is associated with approximately 6% additional weight on the peer, and objectively better peer posteriors receive average weight 0.095 compared with 0.043 for worse peers. Participants therefore do not allocate social influence randomly: relative confidence is informative about which inference is better and helps direct greater influence toward better estimates.

This distinction also clarifies why individual confidence is not enough. The decision to follow another person depends on the comparison between two estimates, not on confidence in one’s own estimate alone. In the baseline data, own cognitive uncertainty by itself is not significantly related to own error, whereas perceived peer uncertainty strongly tracks peer error. What matters for social learning is the relative ranking of the two inferences.

Third, the final block provides supporting evidence for this interpretation. Following Enke and Graeber (2023), participants solve compound belief-updating problems in which signal diagnosticity is uncertain, while the peer solves the corresponding reduced problem. Relative CU rises by 3.47 points and peer weight rises by 0.188. Average gain among disagreements rises from 0.500 points in baseline to 1.905 points in the compound block. More importantly, the relation between peer quality and accuracy gain becomes substantially stronger: the block-by-peer-quality interaction is 0.148 ($p = .001$). When participants rely more heavily on the peer, whether the peer’s inference is actually better becomes correspondingly more important. The same pattern appears in the exactly equivalent subset of compound problems. Because the compound block always follows baseline and changes more than one feature of the task, we use it as reinforcing evidence rather than as the sole identification of the mechanism.

These findings contribute to several strands of experimental and behavioral economics. First, we connect to the literature on social learning. Canonical models emphasize information dispersed across agents (Banerjee, 1992; Bikhchandani et al., 1992; Smith and Sørensen, 2000); Bikhchandani et al. (2024) provide a recent survey. Laboratory studies document information cascades and sys-

tematic departures from Bayesian use of social information (Anderson and Holt, 1997; Çelen and Kariv, 2004; Weizsäcker, 2010; Eyster and Rabin, 2010; Angrisani et al., 2021; De Filippis et al., 2022), while network experiments show related frictions in aggregating distributed signals (Chandrasekhar et al., 2020; Grimm and Mengel, 2020). Our design takes a different limit: participant and peer receive the same prior, likelihood, and realized signal. With perfect Bayesian reasoning, the two posteriors would coincide; more generally, common-prior posteriors cannot remain common-knowledge disagreements (Aumann, 1976). Here disagreement instead reflects heterogeneous inference from common evidence.

The closest precedents are Offerman and Sonnemans (1998) and Agranov et al. (2026), which establish that social observation can be useful for inference, even when others do not possess additional raw information. We build on that insight by moving to a continuous posterior-inference environment with a known Bayesian benchmark. This lets us rank the participant’s and peer’s computations trial by trial, measure how much influence the peer receives, and ask whether relative confidence identifies when that influence is likely to be useful.

Second, the paper connects to evidence on imperfect probabilistic inference and disagreement. A large experimental literature documents systematic departures from Bayesian updating and strong effects of task structure and complexity (Grether, 1980; Charness and Levin, 2005; Benjamin, 2019; Enke and Zimmermann, 2019). Andreoni and Mylovanov (2012) show that disagreement can persist even when subjects receive enough information to agree and highlight beliefs about others’ decision-making skill. We take heterogeneous inference quality as the starting point and ask whether people can recognize a better computation of common evidence well enough to benefit from it.

Third, the paper relates to advice taking, expertise, and the allocation of influence. Decision-makers typically retain too much weight on their own initial judgment (Harvey and Fischer, 1997; Yaniv and Kleinberger, 2000; Yaniv, 2004), underappreciate the benefits of averaging (Larrick and Soll, 2006; Soll and Larrick, 2009; Soll and Mannes, 2011), and can be less responsive to information discovered by others than to equivalent information discovered themselves (Conlon et al., 2026). Other experiments distinguish advice from observational social learning (Schotter, 2003; Çelen et al., 2010), study expert and peer advice (Weizsäcker, 2010; Laple and Barham, 2019; Laple et al., 2023), and show that identifying the more accurate advisor can itself be difficult (Alysandratos et al., 2025). Our design makes advice quality directly observable to the researcher as relative inference accuracy from exactly the same evidence. It therefore separates two margins that are often bundled together: identifying which judgment is better and deciding how much weight to place on it. Participants place little weight on peers overall, but that limited weight is systematically tilted toward better inferences.

Finally, the paper connects social learning to confidence, cognitive uncertainty, and calibration. Advisor confidence can affect trust and advice following (Snizek and Van Swol, 2001), while cognitive uncertainty measures subjective uncertainty about one’s reasoning and predicts heterogeneity in belief and choice errors (Enke and Graeber, 2023). More generally, confidence is useful only when

it tracks performance (Amelio, 2023; Enke et al., 2023), and confidence- or competence-weighted aggregation performs well only when subjective competence is sufficiently calibrated (Koriat, 2012, 2015; Goedde-Menke et al., 2025). Pescetelli et al. (2021) further show that confidence predicts advice seeking and changes of mind. In much of the work most closely related to our mechanism, however, the confidence or competence signal used to evaluate others is directly expressed or elicited. Our setting is different: two individuals independently solve the same inference problem, but the participant observes only the peer’s answer, not the peer’s confidence. The participant must therefore infer the reliability of the peer’s reasoning from the observed posterior and compare it with the perceived reliability of their own reasoning. We show that this resulting relative-confidence judgment is informative about which of the two inferences is more accurate and helps direct social influence toward the better inference.

The rest of the paper is organized as follows. Section 2 describes the experimental design and measures. Section 3 establishes the gains from observational learning under common information. Section 4 studies the calibration of relative confidence. Section 5 examines how relative confidence shapes the allocation and effectiveness of social influence, including the compound-inference block. Section 6 concludes.

2 Experimental Design and Empirical Strategy

2.1 Task, Samples, and Trial Sequence

Participants solve Bayesian inference problems with a binary state. A deck of 100 cards determines the prior probability that bag A rather than bag B is selected. Bags contain complementary proportions of blue and red balls. After the computer selects a bag, participants observe one or two draws with replacement and report the probability that the selected bag was A . Priors take values 30%, 50%, or 70%, and the probability of drawing the bag’s majority color is 70% or 90%. Combining these parameters with the five possible signal descriptions produces 30 statistical-problem cells. In the baseline block, every cell has a unique Bayesian posterior.

The experiment was conducted on Amazon Mechanical Turk in two phases. Phase 1 generated the peer-answer pools. The final matched Phase-1 sample contains 99 participants. For every prior, diagnosticity, and realized-signal cell, the pool stores the numerical posteriors reported in the corresponding exact-parameter task. Across the 30 cells, the pools contain 564 answers; pool sizes range from 5 to 34, with a median of 17.5.

The paper analyzes Phase 2. The final sample contains 204 participants, each of whom completes six belief-revision trials, yielding 1,224 observations. The first three trials form the baseline block and the final three form the compound block described below. Problem order is randomized within block. Appendix B reports the sample construction, dates, demographics, and full elicitation sequence.

Each trial proceeds as follows. Participant i first reports an initial posterior p_i and an incentivized estimate of their placement among 100 participants solving the same task. The participant then reports cognitive uncertainty about the reasoning underlying p_i . Next, the interface displays a posterior p_j drawn uniformly from the Phase-1 answer pool matching the same prior, diagnosticity, and realized signal. Draws are made with replacement across Phase-2 observations. No information about the matched peer is disclosed to participants.

After observing p_j , participant i reports perceived peer cognitive uncertainty and an incentivized probability that their own initial posterior is more accurate than the displayed peer posterior. Finally, the participant can report a revised posterior p'_i . Probability reports and performance-comparison reports are payoff relevant; cognitive-uncertainty reports are not. In each block, one payoff-relevant decision is selected for a \$1.50 bonus. Participants must pass comprehension checks before proceeding.

The baseline block is a common-information environment: participant and peer face the same prior, signal structure, and realized signal. The peer therefore contributes no private signal. Conditional on the statistical-problem cell, the displayed peer answer is randomly assigned from the realized Phase-1 pool. Appendix C documents the assignment algorithm, pool sizes, balance, and randomization procedure.

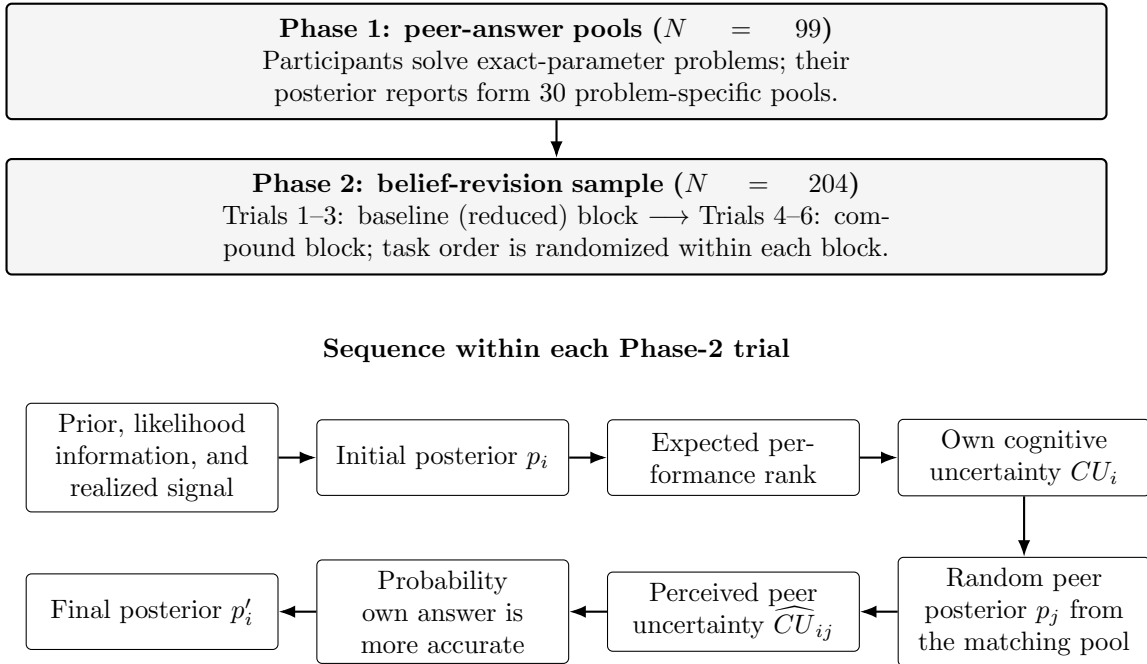


Figure 1: Experimental structure. Phase 1 supplies numerical peer answers for each statistical-problem cell. In Phase 2, an answer is drawn independently from the matching pool on each trial. Participant and peer receive the same prior, diagnosticity, and realized signal in the baseline block. In the compound block, the participant faces a distribution over diagnosticities while the peer answer comes from the corresponding reduced problem.

2.2 Confidence Measures and Outcomes

Cognitive uncertainty (CU) measures subjective uncertainty about the reasoning process that generated an answer (Enke and Graeber, 2023). Participants report CU on a 0–30 slider centered on their probability estimate. Zero means certainty that the optimal posterior equals the stated estimate; larger values produce a wider interval around it. Let CU_i denote own CU and $\widehat{CU}_{ij}$ the uncertainty participant i attributes to peer j . We define relative cognitive uncertainty as

$$R_{ij} = CU_i - \widehat{CU}_{ij}. \quad (1)$$

A larger R_{ij} means that the participant views their own reasoning as less reliable than the peer’s. Own CU is reported before the peer is observed, whereas perceived peer CU is reported afterward. Thus R_{ij} is a relative-confidence judgment formed after seeing the peer answer; it is not an exogenous pre-observation measure.

The second measure is direct. Let $\pi_{ij} \in [0, 100]$ denote the participant’s reported probability that their own initial answer is better than the peer’s. We orient this measure in the same direction as R_{ij} :

$$D_{ij} = 50 - \pi_{ij}. \quad (2)$$

Hence $D_{ij} > 0$ means that the participant regards the peer as more likely to have the better posterior. We call D_{ij} the *direct confidence report*. It is elicited after p_j and perceived peer CU but before the final posterior. The two measures answer related but distinct questions. Relative CU compares perceived uncertainty in the two reasoning processes, whereas the direct report asks which answer is expected to be more accurate. We say that a confidence report is calibrated when greater confidence in an answer predicts lower objective error.

Let p^* denote the Bayesian benchmark. We define own initial error, peer error, and final error as

$$e_i = |p_i - p^*|, \quad (3)$$

$$e_j = |p_j - p^*|, \quad (4)$$

$$e'_i = |p'_i - p^*|. \quad (5)$$

The peer accuracy advantage is

$$q_{ij} = e_i - e_j, \quad (6)$$

so $q_{ij} > 0$ means that the peer posterior is more accurate. The accuracy gain from revision is $g_i = e_i - e'_i$, so positive values mean that the final answer is closer to the Bayesian benchmark.

When $p_j \neq p_i$, weight on the peer is measured by the standard weight-of-advice statistic (Harvey and Fischer, 1997; Yaniv, 2004),

$$WOA_{ij} = \frac{p'_i - p_i}{p_j - p_i}. \quad (7)$$

Our main measure of peer weight is $w_{ij} = \min\{1, \max\{0, WOA_{ij}\}\}$. We also report indicators for any revision and movement toward the peer, absolute movement, winsorized uncapped WOA, and participant-level specifications. Only two baseline observations have negative raw WOA and four exceed one. Thus six of 488 baseline disagreement observations are affected by capping, and the corresponding winsorized-uncapped specification is reported in the appendix.

2.3 Compound Belief-Updating Tasks

The final three trials add uncertainty about the signal-generating process. Following Enke and Graeber (2023), we call the baseline tasks *reduced* belief-updating problems and the later tasks *compound* belief-updating problems. In a reduced problem, the signal diagnosticity is known. In a compound problem, diagnosticity itself is drawn from a known distribution. In our implementation, participants are told that the majority-color share is equally likely to be any integer value from $d_0 - 10$ to $d_0 + 10$, but they do not observe which value was selected. They must therefore update while also integrating over uncertainty about the likelihood of the observed signal.

Enke and Graeber study a closely related reduced-versus-compound comparison in balls-and-urns problems, showing how uncertainty about diagnosticity changes cognitive uncertainty and reported posteriors (Enke and Graeber, 2023). Our design adapts that comparison to a social setting. The participant solves the compound problem, while the displayed peer answer comes from the corresponding reduced problem in which diagnosticity equals the midpoint d_0 . This removes one layer of uncertainty from the peer’s problem and was intended to increase the participant’s uncertainty relative to the peer.

The instructions described the compound and reduced problems as having the same correct answer. This statement is exact for every one-draw problem and for mixed-color two-draw problems. With two same-color draws, the likelihood is nonlinear in diagnosticity, so integrating over the range produces a posterior that differs slightly from the midpoint posterior. We therefore use the literal compound-Bayesian posterior as the main benchmark in this block. Across all compound observations, the mean absolute difference from the midpoint benchmark is 0.16 percentage points and the maximum is 0.97. Appendix I reports the complete calculation and results for the exactly equivalent subset.

The compound block always follows the reduced block and changes both task complexity and the peer’s informational precision. Comparisons across blocks may therefore also reflect order, experience, or fatigue. We use this block as supporting evidence from a setting in which participants report greater uncertainty relative to the peer, not as a causal test of relative CU.

The empirical analysis follows the three-part logic of the paper. Section 3 establishes the gains from observing another inference under common information. Section 4 asks whether relative confidence ranks objective inference quality. Section 5.1 then asks whether participants use this ranking when allocating social influence. Section 5.2 studies the same mechanism in the compound-inference block.

3 Learning from Another Inference under Common Information

We begin with the central outcome: whether the revisions made after observing another posterior improve accuracy. In the baseline block, participant and peer have exactly the same prior, likelihood, and realized signal, so any value of the peer answer comes from heterogeneity in inference rather than additional information.

Result 1. Observing another inference is followed by a significant improvement in accuracy.

Across all 612 baseline observations, final beliefs are 0.415 percentage points closer to the Bayesian benchmark than initial beliefs (clustered SE = 0.155, $p = .008$). Among the 488 observations in which participant and peer initially disagree, the gain is 0.500 points (SE = 0.194, $p = .010$). The improvement is modest in magnitude but statistically clear.

Table 1: Accuracy Gains after Peer Observation

	All Baseline Trials	Initial Disagreements
Mean accuracy gain	0.415*** (0.155)	0.500** (0.194)
Participant clusters	204	200
Observations	612	488

Notes: Accuracy gain is initial absolute error minus final absolute error, in percentage points. Standard errors are clustered by participant. The entries are equivalent to intercept regressions for the mean gain. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 2 shows that the sign of the gain depends on peer quality. When the peer is objectively more accurate, revisions improve accuracy by 1.41 percentage points on average. When the peer is less accurate, revisions reduce accuracy by 0.47 points. This pattern makes clear why common information can still generate useful social learning: the value of the observed posterior depends

on the quality of the second inference.

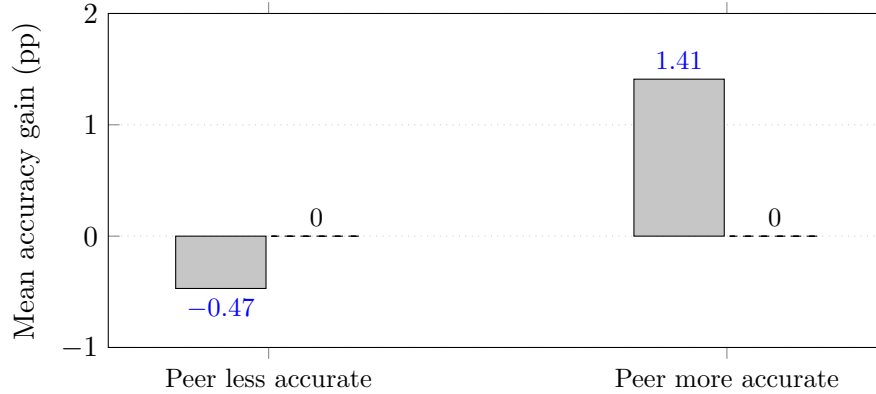


Figure 2: Accuracy changes after observing the peer, by peer quality. Notes: Baseline disagreement observations. Accuracy gain is initial absolute error minus final absolute error; positive values indicate improvement.

The corresponding regression of accuracy gain on an indicator that the peer is more accurate, with participant and problem-by-signal fixed effects, confirms the sharp difference between the two cases. Appendix C reports the continuous peer-quality and randomized-peer reduced-form specifications, Appendix E reports alternative constructions of peer weight, and Appendix H reports the exact-agreement cases. The average before–after improvement is measured over the peer-observation stage and may therefore also incorporate some effect of reconsidering the initial posterior. Such reconsideration appears limited in our setting: when the displayed peer posterior exactly matches the participant’s initial posterior, only 1.65% revise and mean absolute movement is 0.083 percentage points (Appendix H). This pattern is consistent with the immediate condition in Vul and Pashler (2008), where a second estimate elicited immediately after the first provided only modest additional independent information.¹ The randomized assignment of peer posteriors provides separate design-based evidence on how assigned peer quality affects subsequent outcomes (Appendix C).

4 Relative Confidence and Relative Accuracy

The previous section shows that learning is beneficial when influence is directed toward a better inference. We next ask whether participants possess a signal that helps them identify which inference is better.

We define relative cognitive uncertainty as

$$R_{ij} = CU_i - \widehat{CU}_{ij},$$

¹Vul and Pashler (2008) estimate that averaging two immediate guesses had the informational value of approximately 1.11 independent guesses, compared with 1.32 when the second guess was elicited three weeks later.

where higher values indicate that the participant is more uncertain about their own reasoning relative to the peer. We call relative confidence calibrated when this ranking is systematically aligned with objective relative accuracy.

Result 2. Relative cognitive uncertainty is systematically calibrated to relative inference accuracy.

Figure 3 plots the probability that the peer is objectively more accurate across quintiles of relative CU. The probability rises from below one half in the lowest bins to approximately 0.63 in the highest bins. Excluding ties in both relative CU and objective accuracy, the sign of relative CU identifies the better posterior in 59.6% of 440 cases (clustered SE = 2.5 percentage points; $p < .001$ against 50%).

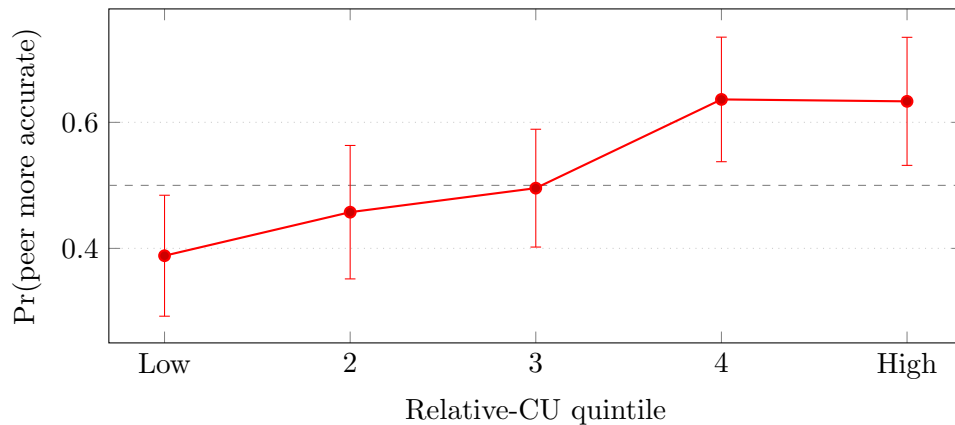


Figure 3: Relative cognitive uncertainty and objective relative accuracy. Notes: Baseline disagreement observations split using global quintile cutpoints of relative CU. Error bars are 95% confidence intervals based on participant-clustered standard errors.

A regression of the peer’s accuracy advantage on relative CU, with participant and problem-by-signal fixed effects, gives a coefficient of 0.525 (SE = 0.230, $p = .022$). Thus relative CU is not merely associated with willingness to revise; it contains information about which of the two initial posteriors is actually closer to the Bayesian benchmark.

Table 2: Relative Confidence and Relative Accuracy

	Peer Accuracy Advantage
Relative CU	0.525** (0.230)
Participant FE	Yes
Problem/signal FE	Yes
Participant clusters	200
Observations	488
R^2	0.620

Notes: Baseline disagreement observations. Peer accuracy advantage is own absolute error minus peer absolute error; positive values mean that the peer is more accurate. Standard errors are clustered by participant. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The social-learning object is relative rather than individual calibration. Own cognitive uncertainty alone does not significantly predict own absolute error in baseline, whereas perceived peer uncertainty strongly predicts peer error. Hence confidence about one’s own reasoning is not sufficient for deciding whether to defer to another person; the economically relevant signal is the ranking of one’s own inference against the peer’s inference. Appendix D reports this decomposition, the direct incentivized comparison measure, answer-feature controls, boundary checks, and cross-validated prediction exercises.

5 Relative Confidence and the Effectiveness of Social Learning

5.1 Relative Confidence and the Allocation of Social Influence

We next ask whether relative confidence is reflected in belief revision. In particular, do participants place more weight on the peer when relative confidence favors the peer, and do more accurate peers receive more weight?

Result 3. Participants direct more social influence toward peers when relative confidence favors the peer.

Figure 4 plots mean peer weight across quintiles of relative CU. Peer weight rises from 0.019 in the lowest quintile to 0.168 in the highest. The corresponding fixed-effects regression yields a coefficient of 0.00594 (SE = 0.00228, $p = .009$): a ten-point increase in relative CU is associated with 0.059 additional weight on the peer, relative to a baseline mean of 0.070.

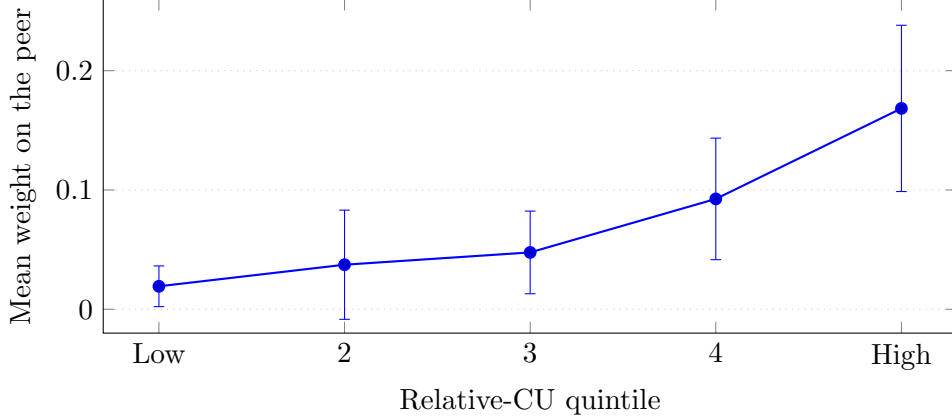


Figure 4: Relative cognitive uncertainty and weight on the peer. Notes: Baseline disagreement observations split using global quintile cutpoints of relative CU. Error bars are 95% confidence intervals based on participant-clustered standard errors.

This allocation is productive on average. Mean peer weight is 0.095 when the peer is objectively more accurate and 0.043 when the peer is less accurate. Regressing peer weight on the peer’s accuracy advantage with participant and problem-by-signal fixed effects yields 0.00182 (SE = 0.00083, $p = .029$). The combination of Results 2 and 3 gives the key mechanism: relative confidence is informative about relative accuracy and is used to direct greater influence toward better inferences.

Table 3: Relative Confidence, Peer Quality, and Social Influence

	(1)	(2)
	Weight on Peer	
Relative CU	0.00594*** (0.00228)	
Peer accuracy advantage		0.00182** (0.00083)
Participant FE	Yes	Yes
Problem/signal FE	Yes	Yes
Dependent-variable mean	0.070	0.070
Participant clusters	200	200
Observations	488	488

Notes: Baseline disagreement observations. Weight on the peer is capped weight of advice. Peer accuracy advantage is own absolute error minus peer absolute error. Standard errors are clustered by participant. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Participants nevertheless use this signal conservatively. Only 10.9% of baseline disagreements move toward the peer, and mean peer weight is 0.07. Leave-one-participant-out weighting rules and within-problem shuffling exercises in Appendix F show that substantially larger gains were available from placing more weight on peers while using relative CU to distinguish where that

weight was most valuable.

5.2 Compound Inference and the Value of Relative Calibration

The final three trials use the compound belief-updating problems described in Section 2. Participants face uncertainty over signal diagnosticity, while the peer answer comes from the corresponding reduced problem. This creates a setting in which participants report substantially greater uncertainty relative to the peer.

Result 4. In the compound block, participants rely more on the peer and learning gains become substantially larger.

Relative CU rises by 3.47 points, the probability of moving toward the peer rises by 20.9 percentage points, and peer weight among disagreements rises by 0.188 (all $p < .001$). At the same time, mean accuracy gain among disagreements rises from 0.500 percentage points in baseline to 1.905 points in the compound block. The within-participant block increase is 1.503 points ($SE = 0.638$, $p = .019$).

Table 4: Compound Block: Reliance and Accuracy Gains

Outcome	Compound-block difference	Standard error
Relative CU	3.47***	
Move toward peer	0.209***	
Weight on peer	0.188***	
Accuracy gain	1.503**	0.638

Notes: Block comparisons use participant fixed effects; the accuracy-gain row refers to disagreement observations and the literal compound-Bayesian benchmark. Exact standard errors and specifications for all outcomes are reported in Appendix I. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 5 summarizes the change in realized learning gains.

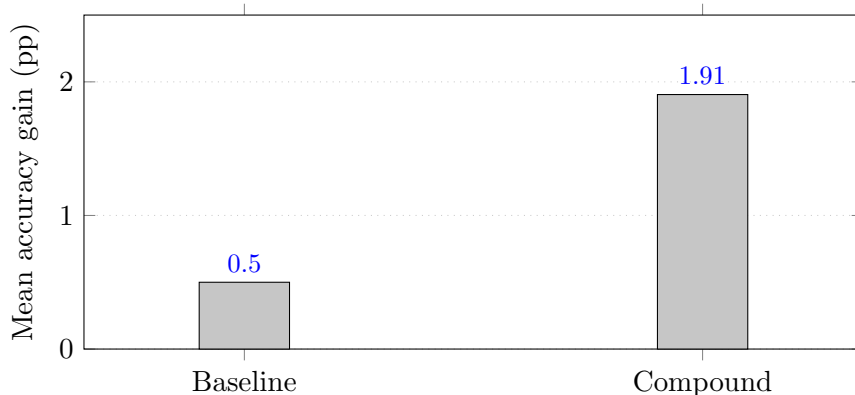


Figure 5: Accuracy gains in baseline and compound-inference disagreements. Notes: The compound block uses the literal compound-Bayesian posterior as the benchmark.

Result 5. When reliance rises, the calibration of relative confidence becomes more consequential for learning outcomes.

Peer quality matters much more in the compound block. The interaction between the compound-block indicator and the peer’s accuracy advantage in a regression of accuracy gain is 0.148 ($SE = 0.046$, $p = .001$). In raw means, participants gain 5.37 points when the peer is more accurate and lose 3.06 points when the peer is less accurate, compared with gains of 1.41 and losses of 0.47 in baseline. Moreover, the interaction between relative CU and peer accuracy advantage in predicting gains is 0.0151 ($p = .001$) in the compound block, compared with 0.0031 ($p = .092$) in baseline.

Figure 6 shows the resulting divergence in gains by peer quality.

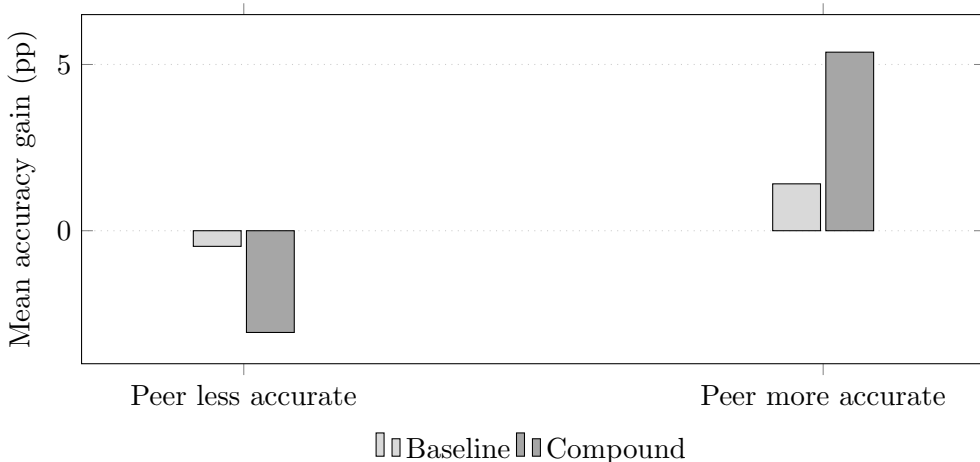


Figure 6: Accuracy gains by peer quality in baseline and compound-inference trials.

The exactly equivalent subset yields the same qualitative pattern: peer weight rises by 0.193 ($p < .001$) and the block-by-peer-quality interaction is 0.214 ($p = .014$). Appendix I reports the exact-equivalence results, one-ball subset, literal-versus-midpoint benchmark audit, and additional specifications. Because the compound block always follows baseline, it is best read as reinforcing evidence for the relative-calibration mechanism rather than as a stand-alone causal estimate of increasing relative CU.

6 Conclusion

People can learn from one another even when they begin with exactly the same statistical information. The reason is simple: common information does not imply common inference. When reasoning is imperfect, two people can produce posterior estimates of different quality from the same evidence, so observing another person’s posterior provides a second computation that can improve accuracy.

Our baseline results establish this possibility in a canonical Bayesian balls-and-urns setting. Final beliefs are more accurate on average after participants observe a peer posterior generated from exactly the same prior, likelihood, and realized signal. The gain is especially clear when the peer is objectively better: revisions improve accuracy when the peer’s inference is superior and reduce accuracy when it is inferior. This extends the common-information insight of Agranov et al. (2026) from binary imitation to continuous posterior inference, where the quality of both computations and the resulting belief revision can be measured directly.

The main contribution is to identify a mechanism behind these gains. Relative cognitive uncertainty is systematically aligned with objective relative accuracy and predicts how much influence the peer receives. Better peer posteriors receive more weight, and the same relative-confidence measure that ranks inference quality also shifts social influence toward the peer. In this sense, relative calibration links metacognition to learning outcomes: the value of confidence in a social setting depends on whether it correctly ranks one’s own inference against another person’s inference.

This comparative perspective is important. Confidence about one’s own reasoning is not sufficient for deciding whether to follow someone else. The relevant decision is inherently relative. In our data, own cognitive uncertainty alone is not significantly calibrated to own error, while perceived peer uncertainty strongly tracks peer error. What supports beneficial learning is therefore not simply confidence, but information about relative reliability. This echoes the broader logic in Amelio (2023): the consequences of social influence depend on the alignment between relative confidence and relative performance.

The compound-inference block reinforces this interpretation. When participants face greater uncertainty about diagnosticity while the peer solves the corresponding reduced problem, relative CU and peer reliance rise sharply. Average learning gains become larger, and peer quality becomes substantially more consequential for final accuracy. When greater reliance is placed on the peer, calibrated relative confidence matters more because following a better inference produces large gains while following a worse inference produces large losses.

Participants nevertheless remain conservative. In baseline, most participants do not revise and average peer weight is small. Counterfactual weighting exercises show that larger gains were available from listening more. This under-reliance is an important quantitative limitation, but it does not overturn the central qualitative result: social influence is not allocated randomly. Participants possess useful information about which inference is better and direct more influence toward better estimates.

The broader implication is that common information does not make social learning redundant. Heterogeneity in reasoning creates a role for observing others, while the calibration of relative confidence helps determine whether that role is beneficial. Social learning can therefore improve inference even without informational asymmetry, provided people can identify whose computation of the common evidence is more reliable.

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A Experimental Instructions and Screens

This section reproduces the instruction and decision screens archived with the original experiment. The screenshots are the original study materials rather than reconstructed illustrations. Numerical parameters and realized signals varied across trials, but the wording and interface were otherwise held fixed within the corresponding task type. The compound block used the same sequence while presenting signal diagnosticity as a range and displaying a peer answer from the corresponding reduced, exact-parameter problem. Table 5 summarizes the elicitation order and payoff relevance of each report.

A.1 Instructions and experimental sequence

Figures 7–10 reproduce the introductory instructions and the two screens used to explain the sequence of a trial. These screens describe the balls-and-urns inference problem and the order in which participants make their own judgment, report confidence-related measures, observe another participant’s answer, and have an opportunity to revise.

Instructions

Please take your time to read the instructions carefully. Your understanding of the instructions will be tested later.

The study is split into two parts.

In this first part, you will be asked to complete **3 similar tasks**.

These tasks can be divided into parts where you **receive relevant information** and parts where you **make decisions**. Importantly, **some** of these decisions may **determine** your **bonus payment**. However, which decision will determine your bonus payment will be picked randomly.

For this reason, you should always **provide your best guess**, to increase the chances that you will receive the bonus payment.

Figure 7: Introductory Instructions: Overview

Task Description

There are **two bags**, bag A and bag B. Each of the bags contains **100 balls**, which may be blue or red. The number of blue and red balls in each bag will vary in each task, but you will be given information on **how many blue and red balls each bag contains**.

Moreover, there is a deck of **100 cards**. Each card may have “A” or “B” written on it.

The number of “A” cards and “B” cards in the deck will vary in each task, but you will be given information on **how many “A” and “B” cards are present in the deck**.

The computer will randomly select a bag, drawing a card from the deck. You will **not observe which bag was selected** this way. Then, **one or more balls** will be drawn from the selected bag. After observing the balls drawn, you will have to provide a **probabilistic guess** on which bag was selected.

The picture below illustrates the process graphically.

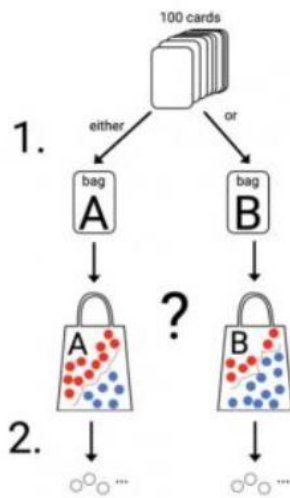


Figure 8: Introductory Instructions: Belief-Updating Task

Timeline

- 1) You are told how many “A” and “B” cards the deck contains and how many blue and red balls bags A and B contain.
- 2) The computer randomly draws a card from the deck. Each card may be drawn with the same probability.
- 3) If the card is an “A card”, the computer draws one or more balls from bag A and if the card is a “B card” the computer draws one or more balls from bag B. The balls are drawn **with replacement**, meaning that every time a ball of any color is drawn it gets replaced in the bag. Hence, the probability of a red or blue ball being drawn depends only on the bag it is drawn from, but not on previous draws.
- 4) You are shown how many blue and red balls were drawn, but not from which bag those were drawn.
- 5) Your task is then to provide a probability between 0% and 100% that the balls were drawn from bag A. Keep in mind that the probability of the sure event is 100% and the probability of the impossible event is 0%. Note that the probability of the balls being drawn from bag B corresponds to 100 minus the probability that the balls were drawn from A (**this decision may determine your bonus payment**).
- 6) After you provide your answer, you will be asked to provide your expectations on your relative performance in this task. For example, you may expect to be in the best 5% or in the worst 10% of the participants, where the performance is measured on the base of how close your answer is to the correct one. To answer, imagine you are competing with other 99 participants that have the same information as you have and try to guess your placement from 1 (first) to 100 (last) (**this decision may determine your bonus payment**).

Figure 9: Experimental Sequence: Part 1

7) You will be asked **how certain** you are about your answer. This point will be explained thoroughly in the next page.

8) You will be shown an **answer from another participant to the exact same task**. This means that the other participant **observed the same ball(s) draw**, with the deck having the same number of **A and B cards** and the A and B bags having the same number of blue and red balls.

9) You will be asked about **how certain you think** the other participant was about the answer you were shown in the previous point.

10) You will be asked **your guess of the probability of your answer being better than the observed answer**. An answer is **better if it is closer** to the correct answer. (this decision may determine your bonus payment).

11) You are given the chance to change your answer after observing the other participant's answer. You may also leave the answer unchanged (this decision may determine your bonus payment).

Please note:

- In each new task the **number of red and blue balls in the bags and the drawn card change**: please think about each of the tasks **independently**.

- As will be clarified in what follows, there is **no real correct answer** to the decision in points 7 and 9. This is why the bonus payment will not depend on that, **so try to answer truthfully**.

Figure 10: Experimental Sequence: Part 2

A.2 Comprehension checks and belief updating

After the instructions, participants answered comprehension questions before proceeding to the decision tasks. Figures 11 and 12 reproduce the archived comprehension screens. Figures 13 and 14 show a representative belief-updating task before and after the signal realization. Participants could review the task parameters while entering their posterior report.

Comprehension questions

The following question test your understanding of the instructions read so far.

Please note that **failing to answer any of the following correctly**, will result in **not being able to proceed further with the study** and **not earning any bonus**.

Which of the following statement is correct?

The number of "A cards" and "B cards" in the deck vary in each task

The probability of bag A being picked in each task is always the same

Which of the following statement is correct?

When asked about the certainty of my guess there is a correct answer

There is no point in giving a strategic answer when asked about the certainty of my guess

Figure 11: Comprehension Questions: Part 1

What is the best way to maximize the chances of receiving the bonus?

Providing any guess, the bonus is received randomly

Providing my best guess for the asked probabilities

What is the best placement among the following?

Top 5%

Top 15%

Top 30%

If you answered the questions correctly, pressing the **right arrow** you will first proceed to an example task, where you will be shown the procedure without being required to provide any answer, and after to the actual task.

Figure 12: Comprehension Questions: Part 2

This task consists in **guessing the probability** of Bags A and B being chosen by the computer.

Number of "**bag A**" cards: 30

Number of "**bag B**" cards: 70

Bag A contains 70 blue balls and 30 red balls.

Bag B contains 30 blue balls and 70 red balls.

Next:

1. The computer **randomly selects one bag** by drawing a card from the deck.
2. The computer **draws 1** ball(s) from the secretly selected bag. Click the arrow for the computer to perform the draw:



Figure 13: Belief-Updating Task Before the Signal Realization

The computer **drew 1 blue ball**.

Please write down **your guess** for **the probability** (between 0 and 100) **of each bag being chosen**.

Probability of bag A :	0
Probability of bag B :	0
Total	0

Please provide you **guess about your placement** in this **specific task** (between 1 and 100):



Figure 14: Belief-Updating Task After the Signal Realization

A.3 Confidence reports and peer evaluation

The cognitive-uncertainty interface is illustrated in Figures 15 and 16. The first shows the response control before a selection and the second after a response has been entered. The experiment also elicited an incentivized population-rank belief before peer observation (Figure 17). After the participant observed the randomly selected peer posterior (Figure 18), the survey elicited perceived peer cognitive uncertainty and the direct pairwise performance assessment described in the main text. Figure 19 reproduces the archived pairwise-comparison screen. The exact direct question was: “Did you perform better than the other participant? Please write your guess of the probability that you did.”

How certain are you that the optimal guess for **bag A probability** is **exactly 20%**?

You are **sure** that the **optimal guess** is **please click the slider**.

Very Certain

Very Uncertain



Figure 15: Cognitive-Uncertainty Elicitation Before a Response

How certain are you that the optimal guess for **bag A probability** is **exactly 20%**?

You are **sure** that the **optimal guess** is **between 11% and 29%**.

Very Certain

Very Uncertain



Figure 16: Cognitive-Uncertainty Elicitation After a Response

The computer **drew 2 blue balls**.

Please write down **your guess** for **the probability** (between 0 and 100) **of each bag being chosen**.

Probability of bag A :	0
Probability of bag B :	0
Total	0

Please provide you **guess about your placement** in this **specific task** (between 1 and 100):

Figure 17: Population-Rank Belief Elicitation

The other participant answered as follows:

Probability of bag A: 10%

Probability of bag B: 90%

Did you perform better than the other participant?

Please write your guess of the probability that you did (100 means that you are sure to have performed better and 0 means that you are sure that you have not).

Figure 18: Display of the Other Participant's Posterior

Did you perform better than the other participant?

Please write your guess of the probability that you did (100 means that you are sure to have performed better and 0 means that you are sure that you have not).

Figure 19: Direct Pairwise Performance Assessment

A.4 Incentives and Phase-1 attention check

Figure 20 reproduces the scoring-rule explanation shown to participants. Probability and comparative-performance reports were payoff relevant, while cognitive-uncertainty reports were not. The Phase-1 data collection used an additional attention-check problem; its archived screen is reproduced in Figure 21. Phase 1 ended after the participant's own reports and did not include the peer-observation and revision stages used in Phase 2.

Bonus payment

*You may receive a bonus of 1.5\$ for this part. Whether or not you receive the bonus depends on the choices you make in the **bonus relevant decision parts**, indicated in the previous screen. One of the 4 relevant decisions will be randomly picked by the computer, with equal probability. The “goodness” of that decision will determine whether you will receive the prize or not, according to the following rules.*

*If the decision picked by the computer is **one of the placement decisions** (points 5 and 10 in the timeline), the computer will randomly draw a number n between 0 and 2500 (each number having the same probability of being drawn).*

*For the decision in point 5, if your actual placement is in the top half of participants, or you will receive the bonus if the square of your stated placement is **lower than n** . On the other hand, if your actual placement is in the lower half of participants, you will receive the bonus if your stated placement is **larger than n** .*

*Similarly, for the decision in point 10, if you performed better than the other participant, you will receive the bonus if the square of your stated probability is **larger than n** and, in the other case, if your stated probability is **lower than n** .*

*If the decision picked by the computer is **one of the probability guesses** (points 6 and 9 in the timeline), the computer will randomly draw a number n between 0 and 2500 (each number having the same probability of being drawn).*

*If in the task the drawn bag was A, if the square of the probability (in percent) you assigned to A is **larger than n** , you receive the bonus. However, if B was drawn, you receive the bonus if the square of the probability assigned to A is **lower than n** .*

All these rules mean that you have the incentive to provide your best guess of your placement or the probabilities, since your best guess improves the chances of receiving the bonus.

Figure 20: Scoring-Rule Instructions

This task consists in **guessing the probability** of Bags A and B being chosen by the computer.

The draw is performed **without replacement**.

Number of **"bag A"** cards: **50**

Number of **"bag B"** cards: **50**

Bag A contains **99 blue balls** and **1 red balls**.

Bag B contains **1 blue balls** and **99 red balls**.

Next:

1. The computer **randomly selects one bag** by drawing a card from the deck.
2. The computer **draws 2 ball(s)** from the secretly selected bag. Click the arrow for the computer to perform the draw:

The computer **drew 2 blue balls**.

Figure 21: Phase-1 Attention-Check Task

B Experimental Procedures and Sample Construction

B.1 Elicitation sequence and incentives

Table 5 records the order of elicitation in each Phase-2 trial. The population-rank belief is elicited before peer observation and asks participants to place their own performance among 100 participants solving the same task. The direct pairwise comparison is elicited after the peer posterior and perceived peer CU. It asks for the probability that the participant's own initial answer is better than the displayed answer. These are distinct variables. The direct question was: "Did you perform better than the other participant? Please write your guess of the probability that you did."

Table 5: Phase-2 Trial Sequence

Order	Event or report	Timing relative to peer	Payoff relevant
1	Initial posterior p_i	Before	Yes
2	Population-rank belief	Before	Yes
3	Own cognitive uncertainty CU_i	Before	No
4	Randomly selected peer posterior p_j	Displayed	—
5	Perceived peer cognitive uncertainty $\widehat{CU}_{ij}$	After	No
6	Probability own answer is better, π_{ij}	After	Yes
7	Revised posterior p'_i	After	Yes

Notes: One payoff-relevant decision in each three-trial block was selected for a \$1.50 bonus. Participants completed comprehension checks before the decision tasks.

B.2 Sample construction

Both phases were conducted on Amazon Mechanical Turk. Table 6 reconstructs the final samples from the archived Qualtrics and approval records. The final Phase-2 sample contains every worker marked completed in the archived approval file after retaining the latest response for duplicate worker IDs. This reproducible rule yields 204 participants; that archived matched sample, rather than a count appearing in earlier draft versions, defines the analysis population used throughout the paper.

Table 6: Sample Construction

Phase	Processing stage	Count
Phase 1	Rows in archived Qualtrics export	187
	Rows passing archived completion marker	171
	Workers marked accepted	99
	Final matched participants	99
Phase 2	Rows in archived Qualtrics export	620
	Rows with usable worker ID	618
	Duplicate response rows removed	138
	Unique workers after retaining latest response	480
	Workers marked completed	204
	Final matched participants	204
	Final trial observations	1,224

Notes: Phase 1 was collected from November 23 to December 8, 2020. Final Phase-2 records run from December 15, 2020 to January 22, 2021. Mean Phase-2 age is 36.5 years (SD 9.7), and mean completion time is 22.8 minutes. Education categories are high school (75), bachelor’s degree (104), master’s degree (22), and Ph.D. or higher (3).

B.3 Phase-1 answer pools

The 30 pools correspond to three priors, two signal diagnosticities, and five signal realizations. They contain 564 numerical posterior reports in total. Cell sizes range from 5 to 34 and have a median of 17.5. Phase 2 reads the list for the relevant cell and selects an element using a uniform random index. Each Phase-2 draw is made independently, so the same numerical answer may appear more than once. The stored file does not contain source-participant identifiers, and the interface displays only the numerical posterior. Table 7 reports the realized pool sizes and the number of distinct numerical answers in each cell.

Table 7: Phase-1 Peer-Answer Pools

Prior	Diagnosticity	Signal	Pool size	Distinct answers
30	70	1 blue	25	10
30	70	1 blue, 1 red	16	5
30	70	1 red	31	9
30	70	2 blue	5	5
30	70	2 red	17	10
30	90	1 blue	16	7
30	90	1 blue, 1 red	8	6
30	90	1 red	25	12
30	90	2 blue	12	9
30	90	2 red	33	12
50	70	1 blue	21	6
50	70	1 blue, 1 red	22	5
50	70	1 red	26	6
50	70	2 blue	12	7
50	70	2 red	13	7
50	90	1 blue	29	8
50	90	1 blue, 1 red	8	2
50	90	1 red	25	11
50	90	2 blue	14	9
50	90	2 red	18	9
70	70	1 blue	23	7
70	70	1 blue, 1 red	26	7
70	70	1 red	22	8
70	70	2 blue	16	9
70	70	2 red	7	5
70	90	1 blue	34	13

Continued on next page

Table 7 – continued

Prior	Diagnosticity	Signal	Pool size	Distinct answers
70	90	1 blue, 1 red	5	4
70	90	1 red	17	10
70	90	2 blue	24	9
70	90	2 red	14	7

C Random Assignment of Peer Answers

C.1 Assignment balance

The archived survey code establishes random sampling from the matching Phase-1 pool. Table 8 provides an additional balance diagnostic by regressing assigned peer error on pre-peer variables and statistical-problem fixed effects. The joint robust Wald test has $p = .079$. Own initial error is individually borderline at the 5% level; the remaining coefficients are small and imprecise. The assignment algorithm, rather than ex post balance, is the basis for identification.

Table 8: Balance of Randomly Assigned Peer Error

	Assigned Peer Error
Own initial error	0.0709* (0.0363)
Own CU	-0.0752 (0.0718)
Pre-peer population-rank belief	-0.0137 (0.0184)
Initial posterior	0.0182 (0.1415)
Initial posterior squared	0.00031 (0.00133)
Problem/signal FE	Yes
Observations	609
Participant clusters	204
Joint test p -value	0.079

Notes: Baseline observations with nonmissing peer answers. Standard errors clustered by Phase-2 participant in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C.2 Reduced-form and randomization inference

For each outcome, we regress the outcome on assigned peer error, problem/signal fixed effects, own initial error, own CU, the pre-peer population-rank belief, the initial posterior, and its square. Conditional randomization inference permutes the assigned peer-error values within statistical-problem cells while holding the realized multiset fixed. Table 9 reports 2,000-permutation two-sided p -values.

Table 9: Randomized Peer Quality: Additional Outcomes

Outcome	Peer-error coefficient	Clustered p	Randomization p
Perceived peer CU	0.1561	< .001	< .001
Direct confidence report	-0.3432	< .001	< .001
Any revision	-0.00124	0.074	0.190
Move toward peer	-0.00117	0.089	0.200
Absolute movement	-0.0201	0.013	0.104
Final absolute error	0.02857	< .001	0.024
Accuracy gain	-0.02857	< .001	0.021

Notes: $N = 609$. Continuous variables are measured in percentage points. Randomization inference conditions on the realized Phase-1 pools and the realized multiset of assigned answers within each problem/signal cell.

D Additional Evidence on Confidence Reports

D.1 Determinants of perceived peer cognitive uncertainty

Because perceived peer CU is elicited after the peer posterior, it may respond to disagreement with the participant’s answer. Table 10 confirms that absolute disagreement predicts greater perceived peer uncertainty. Conditional on disagreement and own error, peer error remains positively related to perceived peer CU. The measure is therefore partly egocentric but not reducible to disagreement alone.

Table 10: Determinants of Perceived Peer Cognitive Uncertainty

	(1)	(2)	(3)
Absolute participant–peer disagreement	0.138*** (0.040)	0.125*** (0.042)	0.119** (0.047)
Own absolute error		-0.048 (0.041)	-0.043 (0.042)
Peer absolute error		0.076* (0.040)	0.080* (0.041)
Same side of Bayesian benchmark			-0.535 (1.677)
Participant FE	Yes	Yes	Yes
Problem/signal FE	Yes	Yes	Yes
Observations	488	488	488

Notes: Baseline disagreement observations. The dependent variable is perceived peer CU. Standard errors clustered by participant in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D.2 Own error, peer error, and the allocation of weight

Table 11 decomposes objective relative accuracy into own and peer error. In column (1), the signs are consistent with selective reliance: weight on the peer is higher when the participant’s initial error is greater and lower when the peer’s error is greater. The estimates are modest, and only the peer-error coefficient is marginally significant. Column (2) adds both confidence reports. The objective-error coefficients attenuate, while the direct confidence report remains predictive of weight on the peer. These regressions are descriptive because subjects do not observe either error. The randomized-peer analysis in Appendix C provides the corresponding design-based evidence that peer quality shifts perceived reliability and final error.

Table 11: Own Error, Peer Error, and Weight on Peer

	(1)	(2)
	Weight on Peer	
Own initial error	0.00184 (0.00116)	0.00169 (0.00119)
Peer error	-0.00181* (0.00105)	-0.00102 (0.00101)
Relative CU		0.00295 (0.00194)
Direct confidence report		0.00259*** (0.00096)
Participant FE	Yes	Yes
Problem/signal FE	Yes	Yes
Dependent-variable mean	0.070	0.070
Participant clusters	200	200
Observations	488	488
R^2	0.573	0.614

Notes: Baseline disagreement observations. Errors are absolute deviations from the Bayesian benchmark in percentage points. Standard errors clustered by participant in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D.3 Answer-based benchmarks

In the baseline disagreement sample, relative CU correctly ranks the more accurate posterior in 59.6% of non-tied comparisons. A rule selecting the posterior closer to 50 is correct in 23.4% of the same comparisons, and a rule selecting the posterior closer to the prior is correct in 31.6%. Their complements perform well because under-inference toward 50 and the prior is common. When disagreement and simple answer-based indicators are added to the continuous relative-accuracy regression, the relative-CU coefficient falls from 0.525 ($p = .022$) to 0.386 ($p = .085$). Relative CU is therefore informative but is not a sufficient statistic for relative accuracy.

Table 12 reports participant-grouped ten-fold cross-validation. A flexible classifier using full task parameters and both posterior reports nearly reconstructs the relative-error ordering. Adding subjective measures does not improve that near-normative prediction. In contrast, both confidence reports improve prediction of actual movement toward the peer, with the direct confidence report and the joint model performing best.

Table 12: Cross-Validated Prediction Benchmarks

Outcome	Predictors	AUC	Log loss
Peer more accurate	Visible task and answer features	0.914	0.366
	+ relative CU	0.913	0.369
	+ direct confidence report	0.913	0.366
	+ both comparisons	0.912	0.369
Move toward peer	Visible task and answer features	0.507	0.368
	+ relative CU	0.627	0.351
	+ direct confidence report	0.650	0.348
	+ both comparisons	0.676	0.338

Notes: Ten-fold cross-validation grouped by participant. Visible features include the task parameters, participant and peer posterior reports, signed and absolute disagreement, distances from 50, the prior and probability boundaries, signal direction, number of draws, and trial position. The peer-accuracy sample has $N = 484$ after removing objective ties; the movement sample has $N = 488$.

E Sparse Revision and Weight-of-Advice Robustness

Revision is concentrated among a minority of participants. Of 204 Phase-2 participants, 157 never revise in the three baseline trials, 39 revise once, six revise twice, and two revise all three times. Among disagreement observations, 11.3% involve any revision and 10.9% move toward the peer. Median capped WOA is zero. Two raw WOA observations are negative and four exceed one.

Table 13 reports robustness of the joint confidence-report specification. Results are similar with winsorized uncapped WOA and after excluding small disagreements. The direct confidence report robustly predicts the extensive margin and most WOA specifications. Relative CU is more sensitive in the trial-level joint model, but its participant-level mean relationship is positive and significant. Neither measure predicts weight precisely among the 53 observations that already move toward the peer, indicating that their principal role is on the extensive margin.

Table 13: Robustness of Confidence Reports and Social Influence

Outcome or sample	Relative CU	Direct confidence report	N
Capped WOA	0.00354* (0.00199)	0.00258*** (0.00095)	488
Winsorized raw WOA $[-1, 2]$	0.00469* (0.00242)	0.00241** (0.00105)	488
Any revision	0.00329 (0.00304)	0.00425*** (0.00142)	488
Move toward peer	0.00398 (0.00298)	0.00387*** (0.00139)	488
Disagreement ≥ 2.5 pp	0.00368* (0.00213)	0.00268** (0.00104)	474
Disagreement ≥ 5 pp	0.00382* (0.00208)	0.00264** (0.00106)	465
Disagreement ≥ 10 pp	0.00506* (0.00299)	0.00204 (0.00153)	373
Participant-level means	0.00513** (0.00221)	0.00139* (0.00072)	200
Conditional on moving toward peer	0.0111 (0.00771)	0.00163 (0.00263)	53

Notes: Except for participant-level means and the conditional-mover specification, regressions include participant and problem/signal fixed effects and use participant-clustered standard errors. The participant-level specification uses HC1 standard errors. The conditional-mover specification includes problem/signal fixed effects and HC1 standard errors. Leave-one-problem-out coefficients remain between 0.00275 and 0.00453 for relative CU and between 0.00218 and 0.00287 for the direct confidence report. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

F Selection and Weight Benchmarks

F.1 Leave-one-participant-out policies

For each held-out participant, the weights are selected using all other participants by minimizing average absolute error over a grid of convex weights. Table 14 reports performance for the held-out participant. The relative-CU rule uses an average training weight of 0.332 when $R_{ij} \leq 0$ and 1.000 when $R_{ij} > 0$. The direct-confidence rule uses 0.571 when $D_{ij} \leq 0$ and 0.975 when $D_{ij} > 0$. The uniform rule uses 0.640.

Table 14: Leave-One-Participant-Out Policy Benchmarks

Rule	Mean Weight	Mean Gain	SE	Difference
Observed choices	0.070	0.494	0.165	—
Uniform rule	0.640	1.249	0.771	0.756 ($p = .317$)
Relative-CU rule	0.576	1.842	0.673	1.348** ($p = .041$)
Direct-confidence rule	0.654	1.463	0.756	0.969 ($p = .188$)
Oracle	0.516	8.768	0.716	8.275*** ($p < .001$)

Notes: Baseline disagreement observations. Gains are percentage-point reductions in absolute error. Standard errors and two-sided tests are clustered by participant. The final column reports the difference from observed choices. The oracle assigns full weight to the more accurate of the two observed posteriors and is included only as an upper benchmark. The relative-CU rule exceeds the uniform rule by 0.592 points ($p = .120$); it exceeds the direct-confidence rule by 0.379 points ($p = .293$).

F.2 Fixed and shuffled-weight benchmarks

Table 15 reports the earlier allocation benchmarks. Applying the observed mean weight uniformly produces less gain than the observed allocation, and shuffling the empirical weights within problem cells reduces the point estimate further. The corresponding one-sided tests are .093 and .056. These exercises hold the realized participant and peer posteriors fixed.

Table 15: Fixed and Shuffled Peer-Weight Benchmarks

Allocation rule	Mean Accuracy Gain (pp)
Observed allocation	0.494
Fixed weight at observed mean ($\simeq 0.07$)	0.289
Shuffled empirical weights within problem	0.141
Fixed 25% weight	0.894
Fixed 50% weight	1.441

Notes: Baseline disagreement observations. The observed-versus-shuffled one-sided permutation p -value is .056. The observed-versus-fixed-mean one-sided participant-bootstrap p -value is .093. The 25% and 50% rules vary intensity and are not claims about optimal weights.

G Tie and Sample Conventions

Weight-of-advice specifications exclude observations in which the initial participant and peer posteriors are exactly equal because the WOA denominator is zero. These agreement observations remain in all analyses whose outcome is defined for them, including revision and final-error regressions.

Sign-ranking exercises exclude objective-error ties and observations in which the relevant confidence report equals zero. In binary peer-quality summaries, “peer more accurate” means strictly lower absolute error; objective ties are neither treated as peer wins nor included in sign-ranking accuracy. In the cross-fitted sign policies, a zero confidence report is grouped with cases that do not favor the peer. Numerical equality uses a tolerance of 10^{-9} .

H Additional Descriptive Accuracy Results

H.1 Exact agreement as a reconsideration benchmark

When participant and peer report exactly the same baseline posterior, only 1.65% of final reports differ from the initial posterior and mean absolute movement is 0.083 percentage points. When their reports differ, 11.2% involve revision and mean absolute movement is 1.166 points. Participant and problem/signal fixed-effect specifications imply a 9.7 percentage point difference in revision probability ($p = .014$) and a 0.91 point difference in absolute movement ($p = .026$). Agreement is endogenous, so this comparison is descriptive and cannot substitute for a randomized no-peer condition.

H.2 Alignment cells

Among baseline disagreements in which relative CU favors the peer, mean weight on the peer is 0.153 and mean gain is 1.979 points when the peer is actually more accurate. When relative CU favors the peer but the participant is actually more accurate, mean weight on the peer is 0.092 and mean gain is -1.114 points. The continuous relative-CU-by-peer-advantage interaction points in the predicted direction but is imprecisely estimated in baseline (0.0031, SE 0.0018, $p = .092$). These cells illustrate why calibration matters without establishing a separate causal interaction.

I Compound Inference and Bayesian Benchmarks

I.1 Literal compound posterior

In the compound block, the participant is told that the signal parameter is uniformly distributed over integer percentage values in a 20-point range centered on the peer’s exact parameter. The literal Bayesian benchmark integrates the likelihood over that distribution. It equals the mid-point benchmark for one-draw and mixed-color two-draw signals. For same-color two-draw signals, Jensen’s inequality creates small differences. Of 612 compound-block observations, 227 are in cells with a nonzero benchmark difference. The mean absolute difference among all compound-block observations is 0.158 percentage points, the mean among changed observations is 0.425, and the

maximum is 0.971.

Using the literal benchmark, mean compound-block gain is 1.385 points across all observations (SE 0.414, $p < .001$) and 1.905 points among disagreements (SE 0.511, $p < .001$). Relative to baseline, the increases are 0.971 points ($p = .044$) and 1.503 points ($p = .019$), respectively. The block-by-peer-advantage interaction is 0.148 (SE 0.046, $p = .001$).

I.2 Exactly equivalent subset

The exactly equivalent subset contains all one-draw and mixed-color two-draw observations, for which the compound and reduced-task posteriors coincide. Table 16 shows that the block strongly raises relative CU, revision, and weight on the peer in this subset. The mean block change in accuracy gain is imprecise, but the interaction between block and peer accuracy advantage remains positive and significant.

Table 16: Compound Block: Exactly Equivalent Subset

Outcome	Block coefficient	SE	p
Own CU	1.822	0.541	$< .001$
Perceived peer CU	-1.902	0.476	$< .001$
Relative CU	3.723	0.657	$< .001$
Any revision	0.220	0.035	$< .001$
Move toward peer	0.205	0.035	$< .001$
Social weight	0.193	0.037	$< .001$
Accuracy gain	0.587	0.730	0.421
Block $\times$ peer accuracy advantage	0.214	0.088	0.014

Notes: One-draw and mixed-color two-draw observations, for which literal and mid-point posteriors coincide exactly. Specifications include participant fixed effects and within-block task-position controls. Social-weight and interaction specifications use disagreement observations.